PROBLEM

1. Hospital Doctor–Shift Assignment – Problem Statement

A hospital needs to assign three doctors, D1, D2, and D3, to three shifts: Morning, Afternoon, and Night, with exactly one doctor assigned to each shift. The assignment must satisfy several constraints: D1 cannot work the Night shift, D2 must be scheduled before D3, D3 cannot work the Morning shift, and every doctor must be assigned exactly one shift. The objective is to use Backtracking Search to systematically explore possible assignments, check the constraints at each step, and find a valid schedule.

2. Robot Grid Navigation – Problem Statement

A robot operates in a 5×5 grid environment and must navigate from a Start (S) position to a Goal (G) position while avoiding obstacles. The robot can move only Up, Down, Left, or Right, and each movement has a cost of 1. The robot must not pass through blocked cells and should use the Manhattan Distance heuristic to guide its search toward the goal. The objective is to demonstrate the search process step by step, including node expansion and priority queue updates, and determine the optimal path and its total cost.

3. Autonomous Rescue Robot – Problem Statement

An autonomous rescue robot is deployed in a disaster-affected building to locate a survivor at the Goal (G), starting from position S. The building is represented as a grid containing blocked cells and risky zones, and the robot can move only Up, Down, Left, or Right. Since the robot has limited sensing capability and can observe only adjacent cells, it must make decisions with incomplete information and dynamically update its knowledge as it moves. Each movement has a cost of 1, while entering a risky zone adds an additional cost of 2. The objective is to use online search to help the robot safely navigate the changing environment while minimizing the total movement cost.